01

Modul : Supervised Learning

k-Nearest Neighbor

KK IF - Teknik Informatika- STEI ITB

Inteligensi Buatan
(Artificial Intelligence)

k-Nearest Neighbor

- Supervised Learning
- Instance-Based Classifier (Store all training data)
- Lazy learner

No hypothesis

Unseen data prediction: Find class from similar stored data

Classification (Predict unseen data)

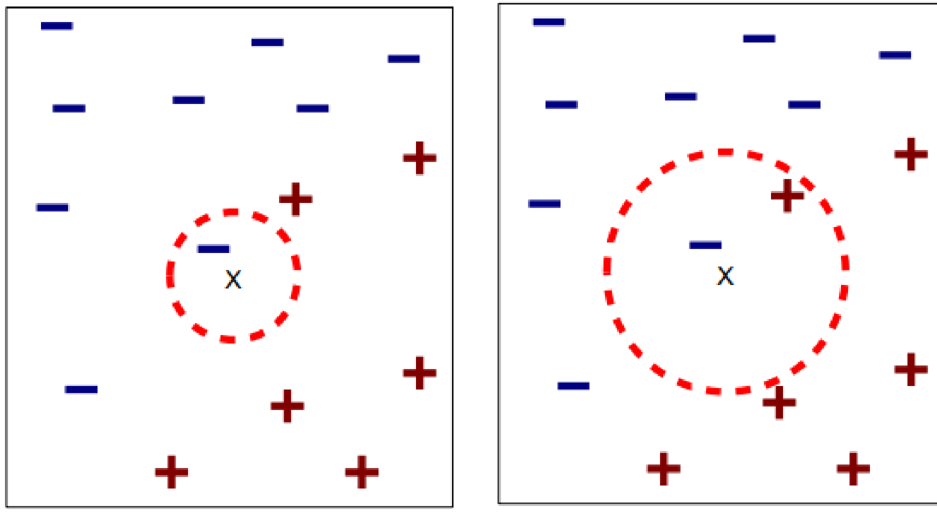
Measures 'distance' of query (unseen data) to all instance (in training data)

- Symbolic attribute: 1 (different value), 0 (same value)
- Numeric attribute: Euclidean Distance

Find k 'most similar' instances (k nearest neighbor)

Find the majority class from k nearest neighbor

Class/ Label Prediction: Majority Class of k nearest neighbor



(a) 1-nearest neighbor (b) 2-nearest neighbor

Example: Play Tennis Dataset

outlook	temp.	humidity	windy	play
sunny	hot	high	false	no
sunny	hot	high	true	no
overcast	hot	high	false	yes
rainy	mild	high	false	yes
rainy	cool	normal	false	yes
rainy	cool	normal	true	no
overcast	cool	normal	true	yes

outlook	temp.	humidity	windy	play
sunny	mild	high	false	no
sunny	cool	normal	false	yes
rainy	mild	normal	false	yes
sunny	mild	normal	true	yes
overcast	mild	high	true	yes
overcast	hot	normal	false	yes
rainy	mild	high	true	no

Classify New Instance: <Sunny, Cool, High, True>

outlook	temp.	humidity	windy	play	Distance
sunny	hot	high	false	no	2
sunny	hot	high	true	no	1
overcast	hot	high	false	yes	3
rainy	mild	high	false	yes	3
rainy	cool	normal	false	yes	3
rainy	cool	normal	true	no	2
overcast	cool	normal	true	yes	2
sunny	mild	high	false	no	2
sunny	cool	normal	false	yes	2
rainy	mild	normal	false	yes	4
sunny	mild	normal	true	yes	2
overcast	mild	high	true	yes	4
overcast	hot	normal	false	yes	2
rainy	mild	high	true	no	2

k = 1 → D-2 → Play: No

k = 2 → D-1, D-2, → Play: No

k = 3 → D-1, D-2, D-6 → Play: No

Notes on k-Nearest Neigbbor

Advantages

- Approximation can be less complex for complex target function

Disadvantages

- Cost of classifying new instance high
- Consider all features → target function depends only on a few features

id	hobi	umur	pendidikan	kelas
1	Game	remaja	sma	1
2	Game	dewasa	s1	2
3	Game	dewasa	diploma	3
4	Baca	dewasa	s1	3
5	Olahraga	dewasa-muda	s1	2
6	Olahraga	dewasa-muda	diploma	3
7	Game	dewasa-muda	sma	1
8	Olahraga	dewasa-muda	sma	1
9	Baca	dewasa-muda	sma	1
10	Game	dewasa-muda	s1	3
11	Baca	Remaja	diploma	1
12	Game	remaja	diploma	2
13	game	dewasa	sma	3

Id	Jarak thd data baru
1	1+1+1=3
2	**1+0+0=1**
3	**1+0+1=2**
4	**1+0+0=1**
5	**0+1+0=1**
6	**0+1+1=2**
7	1+1+1=3
8	0+1+1=2
9	1+1+1=3
10	1+1+0=2
11	1+1+1=3
12	1+1+1=3
13	1+0+1=2

1. Hitung Jarak setiap instance data latih utk data uji berikut:
   hobi = olahraga; umur = dewasa; pendidikan = s1
2. Untuk k=5, maka kelas dari data uji di atas adalah: …

Distance Measurement on Numeric Attributes

$$D = \left(\sum_{i=1}^{n} |p_i - q_i|^p \right)^{1/p}$$

Minkowski distance

$$D_m = \sum_{i=1}^{n} |p_i - q_i|$$

Manhattan distance

$$D_e = \left(\sum_{i=1}^{n} (p_i - q_i)^2 \right)^{1/2}$$

Euclidean distance

Brightness	Saturation	Class
40	20	Red
50	50	Blue
60	90	Blue
10	25	Red
70	70	Blue
60	10	Red
25	80	Blue

Brightness	Saturation	Class
20	35	?

d1 = √(20 - 40)² + (35 - 20)²
= √400 + 225
= √625
= 25

d2 = √(20 - 50)² + (35 - 50)²
= √900 + 225
= √1125
= 33.54

$$D_e = \left(\sum_{i=1}^{n} (p_i - q_i)^2 \right)^{1/2}$$

Euclidean distance

$$\sqrt{(X_2-X_1)^2+(Y_2-Y_1)^2}$$

- X_2 = New entry's brightness (20).
- X_1= Existing entry's brightness.
- Y_2 = New entry's saturation (35).
- Y_1 = Existing entry's saturation.

Brightness	Saturation	Class
40	20	Red
50	50	Blue
60	90	Blue
10	25	Red
70	70	Blue
60	10	Red
25	80	Blue

Brightness	Saturation	Class
20	35	?

Brightness	Saturation	Class	Distance
40	20	Red	25
50	50	Blue	33.54
60	90	Blue	68.01
10	25	Red	10
70	70	Blue	61.03
60	10	Red	47.17
25	80	Blue	45

THANK YOU